Vivek R. Krishnan

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Education

2023–2027 **University of Pennsylvania**, Philadelphia, PA
B.A. in Physics and Mathematics, expected May 2027

Research Interests

I am broadly interested in quantum gravity and holography, with a particular interest in the role of quantum information and entanglement in the emergence of spacetime geometry.

Research Experience

Feb 2026 – present **High-energy theory** — University of Pennsylvania
Undergraduate Researcher; senior thesis advisor: Jonathan J. Heckman

Investigating soft photon theorems and their relationship to generalized one-form symmetries in cosmological spacetimes. Derived Schwinger–Keldysh Ward identities for QED correlators and obtained conditional soft-photon consistency relations. Studied photonization and self-dual Maxwell solutions using twistor theory and the Penrose transform. Investigated a de Sitter extension of HQET, including spacetime-dependent velocity projectors, and explored an alternative in-out scattering framework for asymptotic charges.

Mar 2025 – Mar 2026 **Statistical physics and optimal inference** — University of Pennsylvania
Undergraduate Researcher; advisor: Andrea J. Liu

Developed a stochastic-thermodynamic framework for sequential inference using finite-state physical memories. Derived weak-signal diffusion limits for likelihood-ratio updates and a Green–Kubo friction tensor for reversible Markov memory dynamics, yielding a dissipation metric on the probability simplex. Formulated optimal update protocols under accuracy and sampling-time constraints using stochastic control, Feynman–Kac equations, and constrained optimization. Manuscript preparation continues with Adam Gabriel Frim and Andrea J. Liu.

Apr 2024 – Mar 2025 **Computational particle physics** — University of Pennsylvania
Undergraduate Researcher; advisor: Dylan S. Rankin

Developed a vector-quantized variational autoencoder (VQ-VAE) for unsupervised anomaly detection in ATLAS particle-collision data. Trained classifiers directly on quantized latent representations to test signal retention and identify anomalous events without decoder-based reconstruction of the original detector data.

Jan 2022 – Mar 2024 **Computational XFEL research** — Arizona State University
Student Researcher; advisor: Richard A. Kirian

Developed an automated Python pipeline for characterizing gas dynamic virtual nozzles (GDVNs), including a machine-learning classifier for drip–jet regimes (99.3% accuracy). Contributed particle image velocimetry and statistical analyses of liquid microjets for X-ray free-electron laser experiments.

Publications and Manuscripts

Manuscript **V. Krishnan**, A. G. Frim, and A. J. Liu.
"Optimal Statistical Inference Under Thermodynamic Constraints."
Manuscript in preparation.

Nov 2024 K. Karpos, S. Zaare, D. Manatou, R. C. Alvarez, **V. Krishnan**, C. Ottmar, J. Gilletti, A. Pableo, D. Doppler, A. Ansari, R. Nazari, A. Ros, and R. A. Kirian. "Comprehensive characterization of gas dynamic virtual nozzles for x-ray free-electron laser experiments." *Structural Dynamics* **11**, 064302 (2024).
doi:10.1063/4.0000262

Presentations

Mar 2026 **Optimal Finite-Time Statistical Inference**
Contributed oral presentation, APS Global Physics Summit, *Noise-Driven Dynamics in Far-From-Equilibrium Systems I*, Denver, CO.

Advanced Coursework and Independent Study

Graduate physics Quantum Field Theory I–II; Quantum Mechanics I–II; Statistical Mechanics; General Relativity; Electromagnetic Phenomena; Mathematical Methods of Physics; Advanced Topics in Theoretical Physics.

Mathematics Advanced Analysis; Differential Geometry; Geometric Analysis and Topology; Abstract Algebra; Advanced Linear Algebra; Complex Analysis; Partial Differential Equations.

Spring 2026 **Supersymmetry and Supergravity — Independent Study**
With Mirjam Cvetič; $D = 4$, $\mathcal{N} = 1$ supersymmetric Yang-Mills theory, effective Lagrangians, non-renormalizattion theorems, supersymmetry breaking, and introductory supergravity.
String Theory — Reading Group
Readings from Johnson's *D-Branes* and Becker, Becker, and Schwarz's *String Theory and M-Theory.*

Teaching Experience

Teaching Assistant, University of Pennsylvania

Fall 2026 PHYS 5503 — General Relativity

Spring 2027 PHYS 5516 — Electromagnetic Phenomena *(upcoming)*

Honors and Research Funding

2023-2027 **Roy and Diana Vagelos MLS Scholar**
Received $20,000 in summer research funding across 2025 and 2026.

Spring 2026 **College of Arts & Sciences Undergraduate Travel Grant**
Received funding to attend the 2026 APS Global Physics Summit

2023 **National Merit Scholar**
$2,500 National Merit Scholarship

Academic Service

2025–present **Co-President**, Penn Society of Physics Students
Organize talks, graduate student Q&A panels, and social/academic events fostering interaction within Penn's physics community.

2026–present **Research Peer Advisor**, Center for Undergraduate Research and Fellowships
Advise a cohort of first- and second-year students on identifying research opportunities, contacting faculty, and selecting projects.

2023-present **Problem Writer**, Penn Math Contest
Contribute problems for the inaugural competition serving high school students across the Greater Philadelphia area.